



Qualification Pack



Concrete Construction Work (WorldSkills)

QP Code: CON/Q4105

Version: 1.0

NSQF Level: 4.5

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CON/Q4105: Concrete Construction Work (WorldSkills)

Brief Job Description

A Concrete Construction Worker generally works on commercial and residential projects. There is a direct relationship between the nature and quality of the product required and the payment made by the customer. Therefore, the Concrete Construction Worker has a continuing responsibility to work professionally in order to meet the requirements of the customer and thus maintain and grow the business.

Personal Attributes

The person should be physically fit to work in varying locations and conditions. The person should be organized, diligent, methodical, safety-conscious, and a prompt decision-maker. In addition to being a team player, the individual should have good communication skills.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [CON/N4101: Work Organization, Management and Communication](#)
2. [CON/N4103: Interpretation of Drawings](#)
3. [CON/N4104: Setting out and Measurement](#)
4. [CON/N4110: Concrete Construction Work \(WorldSkills\)](#)
5. [CON/N4105: Quality Check and Handover](#)

Qualification Pack (QP) Parameters

Sector	Construction
Sub-Sector	Real Estate and Infrastructure construction
Occupation	For Skill Competitions
Country	India
NSQF Level	4.5
Credits	17



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Aligned to NCO/ISCO/ISIC Code	NCO-2015/7122.0100
Minimum Educational Qualification & Experience	Basic Literacy and Numeracy (The Competitors must not be older than 22 years in the year of the Competition.)
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	14 Years
Last Reviewed On	NA
Next Review Date	31/01/2026
NSQC Approval Date	31/01/2024
Version	1.0
Reference code on NQR	QG-4.5-CO-01793-2024-V1-CSDCI
NQR Version	1.0

Remarks:

i) There is no specified minimum job entry age for the skill competitions, however, the competitor(s) must not be older than 22 years in the year of the competition as per WorldSkills Standards. ii) The validity of these qualifications shall be only for two years or as per the WorldSkills competition cycle, whichever is earlier". (Please refer clause 4.1.(d) of NCVET Order No. 32001/06/2023/NCVET, dated 26.01.2024)



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CON/N4101: Work Organization, Management and Communication

Description

This NOS deals with ensuring safe and healthy working environment at competition area by proper utilization of safety gears and effective communication skill with juries and competitors.

Scope

The scope covers the following :

- Adopt healthy and safe work practices at competition area
- Management of workplace
- Effective communication at competition area

Elements and Performance Criteria

Adopt healthy and safe work practices at competition area

To be competent, the user/individual on the job must be able to:

- PC1.** Explain health and safety legislation, obligations, and documentation
- PC2.** check and ensure the availability of appropriate and usable Personal Protective Equipment (PPE) at the workplace
- PC3.** use the appropriate PPE according to the nature of work and ensure other personnel also do the same
- PC4.** follow and promote the appropriate measures to ensure hygiene in the workplace, e.g.regular cleaning and sanitization
- PC5.** follow the applicable emergency procedures and ensure the personnel also do the same
- PC6.** Learn precautions for the safe use of power and cordless tools
- PC7.** Apply standards and laws relating to security, safety, and hygiene in plastering and drywall systems

Management of workplace

To be competent, the user/individual on the job must be able to:

- PC8.** establish methods for optimal and economically organized construction site. regarding construction plan and equipment, organization and procedures, material management, staffing and timelines
- PC9.** determine the correct calculations and estimates by using principles and methods
- PC10.** store plasterboards and related products safely and securely
- PC11.** be proactive in own continuous professional development in order to keep abreast of methods of working in the construction industry and changing technologies, for example acoustics, sustainability and environmental impact
- PC12.** Work effectively as part of teams
- PC13.** Work effectively with other trades
- PC14.** Establish and maintain high quality standards and working processes
- PC15.** identify problems promptly and manage their resolution



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- PC16.** select, use and store all materials safely
- PC17.** plan work areas to maximize efficiency and maintain the discipline of regular tidying
- PC18.** principles and methods for determine the correct calculations and estimates
- PC19.** understand technical terminology and symbols

Effective communication at competition area

To be competent, the user/individual on the job must be able to:

- PC20.** visualize and translate customer wishes making recommendations which meet/improve their design and budgetary requirements where qualified to do so
- PC21.** Produce a cost and time estimate for customers
- PC22.** introduce related trades to support customer requirements
- PC23.** work effectively in a team to facilitate efficiency/productivity/quality and cost control
- PC24.** recognize the needs of architects and related trades
- PC25.** explain the importance of swiftly resolving misunderstandings and conflicting demands

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Health, hygiene, and safety legislation, obligations, regulations and documentation
- KU2.** principles of working safely with electricity
- KU3.** Accident/first-aid/fire/emergency procedures and reporting
- KU4.** situations when personal protective equipment must be used
- KU5.** purposes, uses, care, maintenance, and storage of all hand and powered tools and equipment together with their safety implications
- KU6.** purposes, uses, care and storage of materials
- KU7.** sustainability measures applying to the use of green materials and recycling
- KU8.** the significance of planning, accuracy, checking, and attention to detail in all working practices
- KU9.** importance of integrity and trustworthiness
- KU10.** value of managing own continuing professional development
- KU11.** value of building and maintaining trust and productive working relationships
- KU12.** importance of swiftly resolving misunderstandings and conflicting demands
- KU13.** need for security for the storage of tools and materials
- KU14.** criteria for being understandable within teams and to non specialists
- KU15.** recognised IT systems and related professional design software

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain the appropriate data and records
- GS2.** read the appropriate reports and literature concerning the test project



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- GS3.** communicate professionally with all the stakeholders
- GS4.** listen attentively to understand the information/ instructions being shared and take appropriate action
- GS5.** coordinate with co-workers to achieve the work objectives
- GS6.** plan and execute tasks based on priority
- GS7.** identify possible disruptions to work and take appropriate mitigation measures
- GS8.** take prompt action to deal with workplace emergencies and accidents
- GS9.** evaluate all possible solutions to work-related problems and select the best one
- GS10.** follow the recommended practices for the timely completion of work and achievement of organizational objectives



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Adopt healthy and safe work practices at competition area</i>	10	25	-	-
PC1. Explain health and safety legislation, obligations, and documentation	-	-	-	-
PC2. check and ensure the availability of appropriate and usable Personal Protective Equipment (PPE) at the workplace	-	-	-	-
PC3. use the appropriate PPE according to the nature of work and ensure other personnel also do the same	-	-	-	-
PC4. follow and promote the appropriate measures to ensure hygiene in the workplace, e.g.regular cleaning and sanitization	-	-	-	-
PC5. follow the applicable emergency procedures and ensure the personnel also do the same	-	-	-	-
PC6. Learn precautions for the safe use of power and cordless tools	-	-	-	-
PC7. Apply standards and laws relating to security, safety, and hygiene in plastering and drywall systems	-	-	-	-
<i>Management of workplace</i>	10	25	-	-
PC8. establish methods for optimal and economically organized construction site. regarding construction plan and equipment, organization and procedures, material management, staffing and timelines	-	-	-	-
PC9. determine the correct calculations and estimates by using principles and methods	-	-	-	-
PC10. store plasterboards and related products safely and securely	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC11. be proactive in own continuous professional development in order to keep abreast of methods of working in the construction industry and changing technologies, for example acoustics, sustainability and environmental impact	-	-	-	-
PC12. Work effectively as part of teams	-	-	-	-
PC13. Work effectively with other trades	-	-	-	-
PC14. Establish and maintain high quality standards and working processes	-	-	-	-
PC15. identify problems promptly and manage their resolution	-	-	-	-
PC16. select, use and store all materials safely	-	-	-	-
PC17. plan work areas to maximize efficiency and maintain the discipline of regular tidying	-	-	-	-
PC18. principles and methods for determine the correct calculations and estimates	-	-	-	-
PC19. understand technical terminology and symbols	-	-	-	-
<i>Effective communication at competition area</i>	10	20	-	-
PC20. visualize and translate customer wishes making recommendations which meet/improve their design and budgetary requirements where qualified to do so	-	-	-	-
PC21. Produce a cost and time estimate for customers	-	-	-	-
PC22. introduce related trades to support customer requirements	-	-	-	-
PC23. work effectively in a team to facilitate efficiency/productivity/quality and cost control	-	-	-	-
PC24. recognize the needs of architects and related trades	-	-	-	-
PC25. explain the importance of swiftly resolving misunderstandings and conflicting demands	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	70	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	CON/N4101
NOS Name	Work Organization, Management and Communication
Sector	Construction
Sub-Sector	Real Estate and Infrastructure construction
Occupation	For Skill Competitions
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2026
NSQC Clearance Date	31/01/2024



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CON/N4103: Interpretation of Drawings

Description

This NOS deals with understanding of the test project drawings along with its process of measurements.

Scope

The scope covers the following :

- Interpretation and execution of drawings as per standards

Elements and Performance Criteria

Interpretation and execution of drawings as per standards

To be competent, the user/individual on the job must be able to:

- PC1.** Prepare site measurement drawings
- PC2.** Accurately interpret all plans, elevations, sections and enlarged details
- PC3.** Produce basic outline drawings (hand and CAD) including elevations, plans, and sections to full size
- PC4.** produce accurate complex drawings on wood to make figure on the wall/floor
- PC5.** Identify drawing errors or items that require clarification
- PC6.** Determine and check quantities of materials required
- PC7.** Calculate a cost and price for the work
- PC8.** Identify horizontal and vertical key dimensions and all angles
- PC9.** Identify curved work and mortar joint finishes
- PC10.** Interpret all project features and their required construction methods
- PC11.** Establish any features that need special equipment or templates and source these
- PC12.** Recognize specified bonding patterns and obey bonding rules during construction
- PC13.** Provide advice and guidance to other professionals such as architects and quantity surveyors
- PC14.** Keep essential notes on each installation process
- PC15.** Explain complex specialist and technical information about installations to clients and other professionals
- PC16.** Apply mathematic geometry principles to the calculation of angles, areas, perimeters, curves, arcs, volumes, ratios, etc.
- PC17.** Prepare the materials requirements, taking into account increased requirements due to compression, wastage, breakage, etc.
- PC18.** Calculate formwork surfaces and materials requirements
- PC19.** Calculate formwork surfaces and materials requirements for face concrete formwork
- PC20.** Interpret, analyse, and work with construction plans (e.g. design plans, formwork plans, reinforcement plans, detail drawings, etc.) and material and parts lists

Knowledge and Understanding (KU)



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The individual on the job needs to know and understand:

- KU1.** The essential information required for floor plans in construction drawings including sections, datum levels, wall constructions, material codes, depth dimensions, heights, schedules and specification
- KU2.** Interpretation and execution of drawings to ISO-A or ISO-E standards
- KU3.** The importance of checking for missing information or errors, anticipating problems and resolving in advance of the setting out process
- KU4.** The role and use of geometry
- KU5.** Mathematical processes and problem solving
- KU6.** The range of costs to be included in estimates
- KU7.** Trends in the industry including new materials and construction methods
- KU8.** The demands and specific properties and qualities of different building materials, such as organic and inorganic materials, coating materials, mounting materials, jointing and adhesive material
- KU9.** Primary elements such as floor, wall and ceiling systems and storefronts
- KU10.** How to choose efficient attachments, and the correct material for cement, wood and steel constructions
- KU11.** How to fabricate ceiling elements to correlate with heat insulation systems, soundproofing and fire protection systems and decorative aspects in the design of walls and ceilings
- KU12.** Principles and methods of technical construction
- KU13.** The relationship between the properties and characteristics of building materials and their impact on comfortable living spaces
- KU14.** The variety of building materials, their chemical and physical processes, capillary action, porosity, diffusion, and condensation as well as their implications for recycling
- KU15.** External and internal colour schemes, e.g. monochromatic, analogous, complementary, warm/advancing, contrasting and cool/receding

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work-related notes and records
- GS2.** read the relevant literature to learn about the latest developments in the field of work
- GS3.** communicate clearly and politely with co-workers and clients
- GS4.** listen attentively to understand the information/ instructions being shared by the speaker
- GS5.** plan and prioritize tasks to ensure timely completion
- GS6.** coordinate with co-workers to achieve work objectives
- GS7.** take quick decisions to deal with workplace emergencies/ accidents



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Interpretation and execution of drawings as per standards</i>	30	70	-	-
PC1. Prepare site measurement drawings	-	-	-	-
PC2. Accurately interpret all plans, elevations, sections and enlarged details	-	-	-	-
PC3. Produce basic outline drawings (hand and CAD) including elevations, plans, and sections to full size	-	-	-	-
PC4. produce accurate complex drawings on wood to make figure on the wall/floor	-	-	-	-
PC5. Identify drawing errors or items that require clarification	-	-	-	-
PC6. Determine and check quantities of materials required	-	-	-	-
PC7. Calculate a cost and price for the work	-	-	-	-
PC8. Identify horizontal and vertical key dimensions and all angles	-	-	-	-
PC9. Identify curved work and mortar joint finishes	-	-	-	-
PC10. Interpret all project features and their required construction methods	-	-	-	-
PC11. Establish any features that need special equipment or templates and source these	-	-	-	-
PC12. Recognize specified bonding patterns and obey bonding rules during construction	-	-	-	-
PC13. Provide advice and guidance to other professionals such as architects and quantity surveyors	-	-	-	-
PC14. Keep essential notes on each installation process	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. Explain complex specialist and technical information about installations to clients and other professionals	-	-	-	-
PC16. Apply mathematic geometry principles to the calculation of angles, areas, perimeters, curves, arcs, volumes, ratios, etc.	-	-	-	-
PC17. Prepare the materials requirements, taking into account increased requirements due to compression, wastage, breakage, etc.	-	-	-	-
PC18. Calculate formwork surfaces and materials requirements	-	-	-	-
PC19. Calculate formwork surfaces and materials requirements for face concrete formwork	-	-	-	-
PC20. Interpret, analyse, and work with construction plans (e.g. design plans, formwork plans, reinforcement plans, detail drawings, etc.) and material and parts lists	-	-	-	-
NOS Total	30	70	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	CON/N4103
NOS Name	Interpretation of Drawings
Sector	Construction
Sub-Sector	Real Estate and Infrastructure construction
Occupation	For Skill Competitions
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2026
NSQC Clearance Date	31/01/2024



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CON/N4104: Setting out and Measurement

Description

This NOS deals with measurements to be setout on competition area as per the drawing

Scope

The scope covers the following :

- Implement techniques to setout the measurement

Elements and Performance Criteria

Implement techniques to setout the measurement

To be competent, the user/individual on the job must be able to:

- PC1.** Check measurements of the wall/floor conform to the drawing specifications
- PC2.** Produce setting out for templates
- PC3.** Visualize and think through projects, identifying potential challenges early and taking the necessary preventative action
- PC4.** Set out the locations, starting points and lines of projects according to plans and specifications
- PC5.** Set out highly technical designs including brick-on-end, brick-on edge, raked/inclined, curved projecting, recessing brickwork, archways, corbelling, decorative bonding, and battered walling
- PC6.** Accurately interpret the dimensions from drawings and ensure the design is set out within a given tolerance
- PC7.** Check all horizontal and vertical angles
- PC8.** Accurately interpret dimensions from drawings and ensure designs are set out within a given tolerances
- PC9.** Produce any templates/building aids that may be helpful when constructing
- PC10.** Set out datum points of reference for projects
- PC11.** Carry out setting out work using the necessary surveying equipment (pocket rule, tape measure, distance meter, set square, level, etc.)
- PC12.** Measure predetermined structures, joints, and materials for the subsequent face concrete surfaces (anchor holes, shuttering frames, board inserts, distribution and alignment of formwork boards, etc.)
- PC13.** Create horizontal levels and measure heights using spirit levels, water level gauges, and optical devices

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:



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- KU1.** The importance of thinking 'top down' to ensure all features can be set out at the start of projects
- KU2.** The implications for businesses/organizations of not setting out correctly
- KU3.** The templates/building aids which may be helpful for construction
- KU4.** Calculations to assist in measurement and checking projects
- KU5.** Geometrical techniques to assist with the project
- KU6.** Methods of setting out horizontal, vertical, raking and curved surfaces forming plain areas, patterns and motifs

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain the appropriate data and records
- GS2.** read the appropriate reports and literature concerning the field of work
- GS3.** communicate professionally with all the stakeholders
- GS4.** listen attentively to understand the information/ instructions being shared and take appropriate action
- GS5.** coordinate with the co-workers to achieve the work objectives
- GS6.** plan and execute tasks based on priority
- GS7.** identify possible disruptions to work and take appropriate mitigation measures
- GS8.** take prompt action to deal with workplace emergencies and accidents
- GS9.** evaluate all possible solutions to work-related problems and select the best one
- GS10.** follow the recommended practices for the timely completion of work and achievement of organizational objectives



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Implement techniques to setout the measurement</i>	30	70	-	-
PC1. Check measurements of the wall/floor conform to the drawing specifications	-	-	-	-
PC2. Produce setting out for templates	-	-	-	-
PC3. Visualize and think through projects, identifying potential challenges early and taking the necessary preventative action	-	-	-	-
PC4. Set out the locations, starting points and lines of projects according to plans and specifications	-	-	-	-
PC5. Set out highly technical designs including brick-on-end, brick-on edge, raked/inclined, curved projecting, recessing brickwork, archways, corbelling, decorative bonding, and battered walling	-	-	-	-
PC6. Accurately interpret the dimensions from drawings and ensure the design is set out within a given tolerance	-	-	-	-
PC7. Check all horizontal and vertical angles	-	-	-	-
PC8. Accurately interpret dimensions from drawings and ensure designs are set out within a given tolerances	-	-	-	-
PC9. Produce any templates/building aids that may be helpful when constructing	-	-	-	-
PC10. Set out datum points of reference for projects	-	-	-	-
PC11. Carry out setting out work using the necessary surveying equipment (pocket rule, tape measure, distance meter, set square, level, etc.)	-	-	-	-
PC12. Measure predetermined structures, joints, and materials for the subsequent face concrete surfaces (anchor holes, shuttering frames, board inserts, distribution and alignment of formwork boards, etc.)	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Create horizontal levels and measure heights using spirit levels, water level gauges, and optical devices	-	-	-	-
NOS Total	30	70	-	-



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National Occupational Standards (NOS) Parameters

NOS Code	CON/N4104
NOS Name	Setting out and Measurement
Sector	Construction
Sub-Sector	Real Estate and Infrastructure construction
Occupation	For Skill Competitions
NSQF Level	4.5
Credits	2
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2026
NSQC Clearance Date	31/01/2024



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CON/N4110: Concrete Construction Work (WorldSkills)

Description

This NOS deals with the construction of test project for Concrete Construction Work Skill.

Scope

The scope covers the following :

- Construction of formwork
- Construction of reinforcement
- Filling of formworks and treatment
- Removal of formworks and reprocessing

Elements and Performance Criteria

Construction of formwork

To be competent, the user/individual on the job must be able to:

- PC1.** Work manually with materials such as wood, metal, and plastic (for separating, reshaping, connecting)
- PC2.** Measure, lay out and cut wood and work with it manually and using machinery
- PC3.** Make simple trestles, working platforms plus auxiliary equipment, set up protective nets and use them in compliance with the relevant regulations
- PC4.** Make and put together every type of formwork
- PC5.** Make supports and reinforcements (concrete pressure)
- PC6.** Make face concrete formwork
- PC7.** Make slits, apertures, openings, and recesses
- PC8.** Move anchors as directed
- PC9.** Make various joints in combination with the appropriate joint sealants (profiles, sealing strips, expansion joint tapes)
- PC10.** Cut to length, bend, interweave, lay, and anchor structural steel according to bending and reinforcement diagrams and in compliance with reinforcement directives (specifically those concerning bending, radius of curvature, end hooks, brackets, distributors, separators, joints, and connection reinforcements)
- PC11.** Prevent the following problems through correct construction
- PC12.** Use scaffolding appropriately and safely and apply health and safety requirements and legislation

Construction of reinforcement

To be competent, the user/individual on the job must be able to:

- PC13.** Make simple trestles, working platforms plus auxiliary equipment, set up protective nets and use them in compliance with the relevant regulations



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- PC14.** Cut to length, bend, interweave, lay, and anchor structural steel according to bending and reinforcement diagrams and in compliance with reinforcement directives (specifically those concerning bending, radius of curvature, end hooks, brackets, distributors, separators, joints, and connection reinforcements)
- PC15.** Prevent the build-up of rust stains on vertical components and of traces of rust caused by reinforcement residues being left on the undersides of horizontal components
- PC16.** Use scaffolding appropriately and safely and apply health and safety requirements and legislation

Filling of formworks and treatment

To be competent, the user/individual on the job must be able to:

- PC17.** Produce unreinforced and reinforced concrete (mix and transport formula concrete = site-mixed concrete)
- PC18.** Order ready-mixed concrete for the site and transport it using concrete pumps, crane buckets, or conveyors
- PC19.** Apply means of separation before concreting, depending on the formwork lining, using high pressure sprays, brushes, cloths or mechanically
- PC20.** Apply concrete in prepared formwork
- PC21.** Compress concrete using various compressors
- PC22.** Process concrete surfaces by smoothing/removing/levelling, using the tools required to do this
- PC23.** Carry out after-treatment of concrete using covers, spray, humidification, use of after-treatment aids, or by leaving fresh concrete in the formwork beyond the stripping times
- PC24.** Prevent incorrect application and compression of concrete

Removal of formworks and reprocessing

To be competent, the user/individual on the job must be able to:

- PC25.** Strip formwork using tools (e.g. formwork bars)
- PC26.** Clean formwork using e.g. water, manual formwork cleaners
- PC27.** Use hazardous cleaners correctly and safely
- PC28.** Care for and maintain system formwork and replace damaged sections
- PC29.** Sort and store all required formwork parts ready for transportation

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** The impact of Health, Safety, and Environment requirements and legislation on projects
- KU2.** How to use and apply tools, equipment, construction machinery, and working aids (e.g. instruments, measuring devices, etc.) in accordance with operating and handling instructions
- KU3.** How to use and handle manual tools such as hammers, saws, planes, etc., to work with materials such as wood, metal, and plastic
- KU4.** How to use and handle machinery such as drills, saws, sanders, etc., to work with materials such as wood, metal, and plastic, in compliance with safety guidelines
- KU5.** Scaffolding requirements



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- KU6.** The individual formwork components such as form lining (plywood, frame elements, screed protection cover), formwork girders, formwork supports, bolts, formwork clamps, and bracing
- KU7.** The components (formwork girders, tubular steel props, supports, bracings, reinforcements, formwork anchors) and materials (wood, steel) for scaffolding
- KU8.** How to make formwork, including erection, bracing, forming recesses, and stripping formwork
- KU9.** Types of formwork
- KU10.** Concrete coverings
- KU11.** The various types of joint (expansion joints, settling joints, construction joints, and dummy joints), what they do and how they are made
- KU12.** Cutting steel and bending steel bars according to standard specifications
- KU13.** Strengthening steel and reinforcement, categories, and types of strengthening steel plus their designations, categorizations, and delivery forms
- KU14.** The need for after-treatment of the concrete (to counter drying-out, temperature differential, frost, leaching, vibrations) using covers, spray, humidification, use of after-treatment aids or by leaving fresh concrete in formwork beyond the stripping times
- KU15.** Face concrete surfaces in terms of porosity, colour consistency, etc.
- KU16.** Stripping times
- KU17.** The cleaning options depending on the formwork material, such as pressurized water, manual formwork cleaning
- KU18.** Health and safety issues and procedures relating to hazardous cleaning materials
- KU19.** Care and maintenance of system formwork (cleaning, maintenance, repairing damaged sections, working with separating agents)

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** write in at least one language, preferably in the local language of the site
- GS2.** read sketches/routine work drawings, instructions provided for the work and various signboards, safety rules, safety tags, exit route information in one or more languages, preferably in the local language of the site
- GS3.** speak in one or more language, preferably one of the local language at the site
- GS4.** communicate orally and effectively with team members
- GS5.** analyze the safety aspects of the workplace
- GS6.** plan work and organize required resource effectively
- GS7.** complete work as per agreed time schedule and quality parameters
- GS8.** resolve any conflict within the teammates
- GS9.** evaluate the complexity of the tasks
- GS10.** identify any violation of safety norms during the work



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Construction of formwork</i>	10	15	-	-
PC1. Work manually with materials such as wood, metal, and plastic (for separating, reshaping, connecting)	-	-	-	-
PC2. Measure, lay out and cut wood and work with it manually and using machinery	-	-	-	-
PC3. Make simple trestles, working platforms plus auxiliary equipment, set up protective nets and use them in compliance with the relevant regulations	-	-	-	-
PC4. Make and put together every type of formwork	-	-	-	-
PC5. Make supports and reinforcements (concrete pressure)	-	-	-	-
PC6. Make face concrete formwork	-	-	-	-
PC7. Make slits, apertures, openings, and recesses	-	-	-	-
PC8. Move anchors as directed	-	-	-	-
PC9. Make various joints in combination with the appropriate joint sealants (profiles, sealing strips, expansion joint tapes)	-	-	-	-
PC10. Cut to length, bend, interweave, lay, and anchor structural steel according to bending and reinforcement diagrams and in compliance with reinforcement directives (specifically those concerning bending, radius of curvature, end hooks, brackets, distributors, separators, joints, and connection reinforcements)	-	-	-	-
PC11. Prevent the following problems through correct construction	-	-	-	-
PC12. Use scaffolding appropriately and safely and apply health and safety requirements and legislation	-	-	-	-
<i>Construction of reinforcement</i>	10	15	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC13. Make simple trestles, working platforms plus auxiliary equipment, set up protective nets and use them in compliance with the relevant regulations	-	-	-	-
PC14. Cut to length, bend, interweave, lay, and anchor structural steel according to bending and reinforcement diagrams and in compliance with reinforcement directives (specifically those concerning bending, radius of curvature, end hooks, brackets, distributors, separators, joints, and connection reinforcements)	-	-	-	-
PC15. Prevent the build-up of rust stains on vertical components and of traces of rust caused by reinforcement residues being left on the undersides of horizontal components	-	-	-	-
PC16. Use scaffolding appropriately and safely and apply health and safety requirements and legislation	-	-	-	-
<i>Filling of formworks and treatment</i>	5	20	-	-
PC17. Produce unreinforced and reinforced concrete (mix and transport formula concrete = site-mixed concrete)	-	-	-	-
PC18. Order ready-mixed concrete for the site and transport it using concrete pumps, crane buckets, or conveyors	-	-	-	-
PC19. Apply means of separation before concreting, depending on the formwork lining, using high pressure sprays, brushes, cloths or mechanically	-	-	-	-
PC20. Apply concrete in prepared formwork	-	-	-	-
PC21. Compress concrete using various compressors	-	-	-	-
PC22. Process concrete surfaces by smoothing/removing/levelling, using the tools required to do this	-	-	-	-
PC23. Carry out after-treatment of concrete using covers, spray, humidification, use of after-treatment aids, or by leaving fresh concrete in the formwork beyond the stripping times	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC24. Prevent incorrect application and compression of concrete	-	-	-	-
<i>Removal of formworks and reprocessing</i>	5	20	-	-
PC25. Strip formwork using tools (e.g. formwork bars)	-	-	-	-
PC26. Clean formwork using e.g. water, manual formwork cleaners	-	-	-	-
PC27. Use hazardous cleaners correctly and safely	-	-	-	-
PC28. Care for and maintain system formwork and replace damaged sections	-	-	-	-
PC29. Sort and store all required formwork parts ready for transportation	-	-	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CON/N4110
NOS Name	Concrete Construction Work (WorldSkills)
Sector	Construction
Sub-Sector	Real Estate and Infrastructure construction
Occupation	For Skill Competitions
NSQF Level	4.5
Credits	9
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2026
NSQC Clearance Date	31/01/2024



Qualification Pack

CON/N4105: Quality Check and Handover

Description

This NOS deals with the quality check of test project prepared as per standards and handing over the same to the concerned juries.

Scope

The scope covers the following :

- Quality check of the test project
- Handover the project to juries

Elements and Performance Criteria

Quality check of the test project

To be competent, the user/individual on the job must be able to:

- PC1.** Inspect equipment, structures, and/or material to identify the nature and causes of errors, defects, or problems
- PC2.** Think critically by using logic and reasoning to identify the strengths and weaknesses of alternative solutions, conclusions or approaches to problems
- PC3.** Identify actual and potential problems
- PC4.** Analyse information and evaluate options to choose and use the best solution
- PC5.** Evaluate solutions and optimize results
- PC6.** The need for all work to be presented to meet customer and related trades needs and expectations
- PC7.** The importance of joint finishing in line with the specifications provided
- PC8.** The different techniques of applying different joint finishes

Handover the project to juries

To be competent, the user/individual on the job must be able to:

- PC9.** understand the process of marking and assessment criteria of respective trade
- PC10.** respond to the juries on the queries being raised by them
- PC11.** Make and follow through decision
- PC12.** Apply appropriate techniques according to buildings' history and fabrication, whilst maintaining their integrity for both internal and external surfaces
- PC13.** Leave work areas in a suitable condition for inspection and subsequent work
- PC14.** Report positive and negative variances in work processes and results, together with their implications
- PC15.** Organize any waste material so that it can be disposed of or recycled efficiently

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:



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- KU1.** The need for all work to be presented to meet customer and related trades needs and expectations
- KU2.** The importance of joint finishing in line with the specifications provided
- KU3.** Presentation including the brushing and cleaning of brickwork plus the tidying and cleaning of the work area (specific skill)
- KU4.** The different techniques of applying different joint finishes
- KU5.** The required quality standards for the task in hand
- KU6.** The nature and causes of substandard work and defects
- KU7.** The available range of quality checks and methods Alternative methods for effecting remedies and repairs

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** maintain work-related notes and records
- GS2.** read the relevant literature to learn about the latest developments in the field of work
- GS3.** listen attentively to understand the information/ instructions being shared by the speaker
- GS4.** communicate clearly and politely with co-workers and clients
- GS5.** coordinate with co-workers to achieve work objectives
- GS6.** plan and prioritize tasks to ensure timely completion
- GS7.** identify possible disruptions to work and take appropriate preventive measures
- GS8.** take quick decisions to deal with workplace emergencies/ accidents
- GS9.** evaluate all possible solutions to a problem to select the best one



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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Quality check of the test project</i>	20	48	-	-
PC1. Inspect equipment, structures, and/or material to identify the nature and causes of errors, defects, or problems	-	-	-	-
PC2. Think critically by using logic and reasoning to identify the strengths and weaknesses of alternative solutions, conclusions or approaches to problems	-	-	-	-
PC3. Identify actual and potential problems	-	-	-	-
PC4. Analyse information and evaluate options to choose and use the best solution	-	-	-	-
PC5. Evaluate solutions and optimize results	-	-	-	-
PC6. The need for all work to be presented to meet customer and related trades needs and expectations	-	-	-	-
PC7. The importance of joint finishing in line with the specifications provided	-	-	-	-
PC8. The different techniques of applying different joint finishes	-	-	-	-
<i>Handover the project to juries</i>	10	20	-	-
PC9. understand the process of marking and assessment criteria of respective trade	-	-	-	-
PC10. respond to the juries on the queries being raised by them	-	-	-	-
PC11. Make and follow through decision	-	-	-	-
PC12. Apply appropriate techniques according to buildings' history and fabrication, whilst maintaining their integrity for both internal and external surfaces	-	-	-	-
PC13. Leave work areas in a suitable condition for inspection and subsequent work	-	-	-	-



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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC14. Report positive and negative variances in work processes and results, together with their implications	-	-	-	-
PC15. Organize any waste material so that it can be disposed of or recycled efficiently	-	-	-	-
NOS Total	30	70	-	-



Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	CON/N4105
NOS Name	Quality Check and Handover
Sector	Construction
Sub-Sector	Real Estate and Infrastructure construction
Occupation	For Skill Competitions
NSQF Level	4.5
Credits	3
Version	1.0
Last Reviewed Date	31/01/2024
Next Review Date	31/01/2026
NSQC Clearance Date	31/01/2024

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

For assessment Experts use specific measurement points which are marked on the drawings. Measurement tools are used to take measurements at these points. Each module/task/section is completed on the assigned day so that progressive marking can take place. The marking of modules will start when all Competitors have finished their module. To ensure transparency, each Competitor is provided the same Mark Summary Form as used by the Experts. A master set of marking tools provided by the Competition Organizer is available during familiarization. All measurements are taken using the master set of measuring equipment. As per the Competition Rules a majority vote (50% +1) is needed to:

- Change the Marking Scheme (within limits specified in the Technical Description);
- Change competition sequence or content;
- Agree on a solution for disputes concerning marks awarded etc.
- The assessment information is not provided to the marking team until all Competitors have completed the work that is assessed.
- The panel of Experts will assist the Skill Competition Manager and the Chief Expert to select the Experts



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for judgement marking.

- The selected Experts possess the suitable industry and competition experience to fulfil the role.
- Four Experts are used in the judgement marking team
- 0-3 Scale:

□ 0: performance below industry standard

□ 1: performance meets industry standard

□ 2: performance meets and, in specific respects, exceeds industry standard

□ 3: performance wholly exceeds industry standard and is judged as excellent

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Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
CON/N4101.Work Organization, Management and Communication	30	70	0	0	100	10
CON/N4103.Interpretation of Drawings	30	70	0	0	100	10
CON/N4104.Setting out and Measurement	30	70	0	0	100	15
CON/N4110.Concrete Construction Work (WorldSkills)	30	70	0	0	100	50
CON/N4105.Quality Check and Handover	30	70	0	0	100	15



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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
Total	150	350	-	-	500	100



Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training



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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.



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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.